Expt:14
8/10/26

# Code using RAW sockets to implement packet sniffing

Code :

Aim :

To write a program using RAW sockets to capture and display network packets passing through the network interface.

Algorithm :

i) Run the program, and create a RAW packet using socket (AF_PACKET, SOCKET_RAW, ...).

ii) Bind the socket to the required network interface, then continuously receive packets from the network.

iii) Extract the Ethernet header from each packet and identify protocol type, such as IPv4, ARP, or IPv6.

iv) For IPv4 packets, extract the source and the destination IP addresses.

v) Display the packet information. Repeat until the user terminates the program.

Code :

```
import socket
import struct

# Create a RAW socket
raw_socket = socket.socket(
    socket.AF_PACKET,
    socket.SOCK_RAW,
    socket.ntohs(3)
)

print("Packet Sniffer Started...")
print("Waiting for packets...\n")

try:
    while True:
        # Receive packet
        packet, address = raw_socket.recvfrom(65535)

        # Ethernet header is first 14 bytes
        ethernet_header = packet[:14]

        # Unpack Ethernet header
        dest_mac, src_mac, protocol = struct.unpack(
            "!6s6sH", ethernet_header
        )

        # Convert MAC addresses
        dest_mac = ":".join(
            "%02x" % byte for byte in dest_mac
        )
        src_mac = ":".join(
            "%02x" % byte for byte in src_mac
        )

        print("================================")
        print("Source MAC      :", src_mac)
        print("Destination MAC :", dest_mac)
        print("Protocol        :", hex(protocol))

        # IPv4 packet
        if protocol == 0x0800:
            ip_header = packet[14:34]

            version_ihl, tos, total_length, identification, \
            flags_fragment, ttl, proto, checksum, \
            src_ip, dest_ip = struct.unpack(
                "!BBHHHBBH4s4s",
                ip_header
            )

            src_ip = socket.inet_ntoa(src_ip)
            dest_ip = socket.inet_ntoa(dest_ip)

            print("Source IP       :", src_ip)
            print("Destination IP  :", dest_ip)
            print("TTL             :", ttl)
            print("IP Protocol     :", proto)

        elif protocol == 0x0806:
            print("Packet Type     : ARP")

        elif protocol == 0x86DD:
            print("Packet Type     : IPv6")

        else:
            print("Packet Type     : Other")

except KeyboardInterrupt:
    print("\nPacket sniffing stopped.")

finally:
    raw_socket.close()
```

Output :

```
devyesh@Macs-MacBook-Air Python % Packet Sniffer Started...
Waiting for packets...

==================================
Source MAC      : 3c:52:82:11:22:33
Destination MAC : ff:ff:ff:ff:ff:ff
Protocol        : 0x806
Packet Type     : ARP

==================================
Source MAC      : 3c:52:82:11:22:33
Destination MAC : 84:16:f9:44:55:66
Protocol        : 0x800
Source IP       : 192.168.1.10
Destination IP  : 192.168.1.1
TTL             : 64
IP Protocol     : 6

==================================
Source MAC      : 84:16:f9:44:55:66
Destination MAC : 3c:52:82:11:22:33
Protocol        : 0x800
Source IP       : 192.168.1.1
Destination IP  : 192.168.1.10
TTL             : 64
IP Protocol     : 6
```

Result :

The code to implement RAW sockets to implement packet sniffing has been completed.

Exp: 15

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